

HAO YU

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Department of Computer Science, Boston University, Boston, MA, 02215

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EDUCATION

Boston University, Boston, MA, USA

Sep 2019 - Present

PhD Student in Computer Science

Advisor: Prof. Margrit Betke

GPA: 3.97 / 4.0

Zhejiang University, Hangzhou, Zhejiang, China

Sep 2015 - Jun 2019

B.E. in Computer Science and B.A. in English Language and Literature

GPA: 3.95 / 4.0

RESEARCH PROJECTS

Leveraging Affect Transfer Learning for Behavior Prediction in an Intelligent Tutoring System

January 2021 - May 2021

Supervisor: Prof. Margrit Betke

Graduate researcher at Boston University

- Proposed a video-based transfer learning approach for predicting problem outcomes of students working with an intelligent tutoring system (ITS) by analyzing their faces and gestures.
- Created a large labeled dataset of student interactions with an intelligent online math tutor.
- Our model achieved a 50% relative increase in mean F-score over the previous state-of-the-art method on this new dataset.

Measuring and Integrating Facial Expressions and Gaze as Indicators of Engagement and Affect in Tutoring Systems

September 2020 - January 2021

Supervisor: Prof. Margrit Betke

Graduate researcher at Boston University

- Conducted two studies using computer vision techniques to measure students' engagement and affective states from their head pose and facial expressions, as they use an online tutoring system.
- Presented a Head Pose Tutor, which estimates the real-time head direction of students and responds to potential disengagement, and a Facial Expression-Augmented Teacher Dashboard, that identifies students' affective states and provides enhanced information to teachers.
- Preliminary results on our collected video data indicated promising accuracy in detecting head orientation. A usability study was conducted where the participants rated the system highly in general.

Facial Emotion Analysis for Videos of Presidential Candidates

September 2020 - Present

Supervisor: Prof. Margrit Betke

Graduate researcher at Boston University

- Collected a large video database of YouTube videos and TV news of 2020 US presidential candidates.
- Annotated the videos with emotion labels using crowd sourcing on Amazon Mechanical Turk.
- Proposed a multi-modal system for facial emotion recognition, which uses pretrained ResNet-18 to extract facial features and uses BERT (Bidirectional Encoder Representations from Transformers) to extract text features. A frame attention module was then adopted for video-based recognition.

Fake Face Video Detection Using Talking Profile

January 2019 - April 2019

Supervisor: Prof. Terence Sim

Research Intern at National University of Singapore

- Proposed a detection algorithm for recognizing fake face videos generated by face swapping using autoencoders and generative adversarial networks (GANs).

- Used talking profile which consists of multiple types of facial and body motions to identify deepfakes.
- Trained support vector machines (SVMs) for classifying real or deepfake videos in DeepfakeTIMIT dataset. Achieved promising classification performance.

Hide-and-Seek: A Data Augmentation Technique

Jul 2018 - Sep 2018

Supervisor: Prof. Yong Jae Lee

Research Intern at University of California, Davis

- Applied Hide-and-Seek, a general purpose data augmentation technique on improving performance of CNN models of Emotion Recognition, Age Estimation, Gender Estimation and Person Re-identification.
- Achieved 1% - 5% relative performance improvement by applying Hide-and-Seek for all tasks. Show-cased the advantage of Hide-and-Seek on various visual recognition problems.

PROFESSIONAL ACTIVITIES

Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)

2019

TEACHING EXPERIENCE

Introduction to Computer Science II (CS 112), Teaching Fellow, Boston University

Instructors: Prof. Christine Papadakis-Kanaris and Prof. David G. Sullivan

2020 Spring

Image and Video Computing (CS 585), Teaching Fellow, Boston University

Instructors: Prof. Margrit Betke

2021 Spring

RELEVANT COURSEWORK

- **Graduate Coursework:** Deep Learning, Machine Learning, Big Data Systems for Data Science, Optimization Theory II, Randomness in Computing.
- **Undergraduate Coursework:** Computer Vision, Artificial Intelligence, Information Retrieval.

TECHNICAL STRENGTHS

Computer Languages

Python, C/C++, Java, MATLAB, L^AT_EX, JavaScript

DL Frameworks

PyTorch, Tensorflow, Caffe, Keras

Toolkits

OpenCV, Scikit-learn, SciPy, Pandas, Numpy, CUDA, MySQL, Hadoop

PUBLICATIONS

[1] **Hao Yu**, Ankit Gupta, Will Lee, Ivon Arroyo, Margrit Betke, Danielle Allesio, Tom Murray, John Magee, and Beverly P. Woolf. "Measuring and Integrating Facial Expressions and Head Pose as Indicators of Engagement and Affect in Tutoring Systems." In International Conference on Human-Computer Interaction, pp. 219-233. Springer, Cham, 2021.

[2] Nataniel Ruiz, **Hao Yu**, Mona Jalal, Danielle Allesio, Ajjen Joshi, Tom Murray, Vitaly Ablavsky, John Magee, Jacob Whitehill, Ivon Arroyo, Beverly Woolf, Stan Sclaroff, Margrit Betke. "Leveraging Affect Transfer Learning for Behavior Prediction in an Intelligent Tutoring System." In 2021 16th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2021), IEEE, 2021 (accepted).

[3] Kevin Deglado, Juan Manuel Origgi, Tania Hansapoor, **Hao Yu**, Danielle Allesio, Ivon Arroyo, Will Lee, Margrit Betke, Beverly Woolf, Sarah Adel Bargal. "Student Engagement Dataset." In Proceedings of the IEEE International Conference on Computer Vision (ICCV) workshops, 2021 (accepted).

[3] Krishna Kumar Singh, **Hao Yu**, Aron Sarmasi, Gautam Pradeep, and Yong Jae Lee. "Hide-and-seek: A data augmentation technique for weakly-supervised localization and beyond." arXiv preprint arXiv:1811.02545, 2018.